

Question 1

Write an if/else statement that assigns 0 to x when y is equal to 10. Otherwise assign 1 to x.

```
1  #include <iostream>
2  using namespace std;
3  int main(){
4
5      int x, y;
6      if (y == 10)
7          x = 0;
8      else
9          x = 1;
10 }
```

Question 2

Write an if statement that prints "The number is valid" if temperature is between -50 and 150.

```
1  #include <iostream>
2  using namespace std;
3  int main(){
4      int temperature;
5      cin >> temperature;
6
7      if (temperature >= -50 && temperature <= 150)
8          cout << "The number is valid";
9      }
10
11
12
```

Question 3

Program to input two numbers and display smaller and larger number.

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int a, b;
6      cout << "Enter two numbers: ";
7      cin >> a >> b;
8
9      if (a > b)
10         cout << "Larger: " << a << "\nSmaller: " << b;
11     else if (b > a)
12         cout << "Larger: " << b << "\nSmaller: " << a;
13     else
14         cout << "Both are equal";
15
16     return 0;
17 }

```

Question 4

Create two variables a and b with same value, add 0.2 to a and compare.

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double a = 10, b = 10;
6
7      a = a + 0.2;
8
9      if (a > b)
10         cout << "a is larger, b is smaller";
11     else if (b > a)
12         cout << "b is larger, a is smaller";
13     else
14         cout << "Both are equal";
15
16     return 0;
17 }

```

Question 5

Divide two numbers only if denominator is not zero.

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double num, den;
6
7      cout << "Enter numerator and denominator: ";
8      cin >> num >> den;
9
10     if (den != 0)
11         cout << "Result = " << num / den;
12     else
13         cout << "Division by zero is not possible.";
14
15     return 0;
16 }

```

Question 6

Find True/False for given conditions (a=2, b=4, c=6)

```

1  #include <iostream>
2  using namespace std;
3  int ()
4  {
5      int main
6      Given: a=2, b=4, c=6
7
8      A) a == 4 || b > 2 ?
9      B) 6 <= c && a < 3 ?
10     C) 1 != b && c != 3 ?
11     D) a >= -1 || a <= b ?
12     E) !(a > 2) ?
13 }

```

Question 7

Program to determine 1st, 2nd, 3rd position of runners.

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int r1, r2, r3;
6      cout << "Enter times of 3 runners: ";
7      cin >> r1 >> r2 >> r3;
8
9      if (r1 < r2 && r1 < r3)
10         cout << "Runner 1 is first\n";
11     else if (r2 < r1 && r2 < r3)
12         cout << "Runner 2 is first\n";
13     else
14         cout << "Runner 3 is first\n";
15 }

```

Question 8

Geometry Calculator (Circle, Rectangle, Triangle)

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int choice;
6      cout << "1.Circle\n2.Rectangle\n3.Triangle\nEnter choice: ";
7      cin >> choice;
8
9      if (choice == 1) {
10         double r;
11         cin >> r;
12         cout << "Area = " << 3.14159 * r * r;
13     }
14     else if (choice == 2) {
15         double l, w;
16         cin >> l >> w;
17         cout << "Area = " << l * w;
18     }
19     else if (choice == 3) {
20         double b, h;
21         cin >> b >> h;
22         cout << "Area = " << 0.5 * b * h;
23     }
24 }

```

Question 9

Display ASCII values of alphabets using type casting.

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      char ch;
6      for (ch = 'A'; ch <= 'Z'; ch++)
7          cout << ch << " = " << (int)ch << endl;
8  }

```

Question 10

Calculate weight and check if too heavy or too light.

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double mass, weight;
6      cout << "Enter mass: ";
7      cin >> mass;
8
9      weight = mass * 9.8;
10     cout << "Weight = " << weight << endl;
11
12     if (weight > 1000)
13         cout << "Too heavy";
14     else if (weight < 10)
15         cout << "Too light";
16 }

```

Question 11

Add bonus if sales > 50000

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double sales;
6      if (sales > 50000) {
7          sales = sales + (0.25 * sales) + 250;
8      }
9      cout << sales;
10 }

```

Question 12

Loan qualification program

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double income;
6      int years;
7
8      cin >> income >> years;
9
10     if (income >= 35000 && years > 5)
11         cout << "Qualified for loan";
12     else
13         cout << "Not qualified";
14 }
```

Question 13

TV model features display

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int model;
6      cin >> model;
7
8      if (model == 300)
9          cout << "Picture-in-picture, Stereo sound, Remote control";
10     else if (model == 200)
11         cout << "Stereo sound, Remote control";
12     else if (model == 100)
13         cout << "Remote control only";
14 }
```

Question 14

Magic date program

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int m, d, y;
6      cin >> m >> d >> y;
7
8      if (m * d == y)
9          cout << "MAGIC";
10     else
11         cout << "NOT MAGIC";
12 }

```

Question 15

Commission calculation

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double sales, commission;
6
7      cin >> sales;
8
9      if (sales <= 10000)
10         commission = 0.10;
11     else if (sales <= 15000)
12         commission = 0.15;
13     else
14         commission = 0.20;
15
16     cout << commission;
17 }

```